

Fig. 5.4



Fig. 5.5

A driver took the job of delivering a truck of books ordered on line. The truck is of $3m$ wide and $2.7m$ high, while driving he noticed a sign at the semielliptical entrance of a tunnel; Caution! Tunnel is of $3m$ high at the centre peak. Then he saw another sign; Caution! Tunnel is of $12m$ wide. Will his truck pass through the opening of tunnel's archway? We will be able to answer this question at the end of this chapter.



Fig. 5.6

Learning Objectives

Upon completion of this chapter, students will be able to

- write the equations of circle, parabola, ellipse, hyperbola in standard form,
- find the centre, vertices, foci etc. from the equation of different conics,
- derive the equations of tangent and normal to different conics,
- classify the conics and their degenerate forms,
- form equations of conics in parametric form, and their applications.
- apply conics in various real life situations.

5.2 Circle

The word circle is of Greek origin and reference to circles is found as early as 1700 BC (BCE). In Nature circles would have been observed, such as the Moon, Sun, and **ripples in water**. The circle is the basis for the wheel, which, with related inventions such as gears, makes much of modern machinery possible. In mathematics, the study of the circle has helped to inspire the development of geometry, **astronomy** and calculus. In Bohr-Sommerfeld theory of the atom, **electron orbit** is modelled as circle.

Definition 5.1

A circle is the locus of a point in a plane which moves such that its distance from a fixed point in the plane is always a constant.

The fixed point is called the **centre** and the constant distance is called **radius** of the circle.

5.2.1 Equation of a circle in standard form

(i) Equation of circle with centre (0, 0) and radius r

Let the centre be $C(0,0)$ and radius be r and $P(x,y)$ be the moving point.

Note that the point P having coordinates (x,y) is represented as $P(x,y)$.

Then, $CP = r$ and so $CP^2 = r^2$

Therfore $(x-0)^2+(y-0)^2 = r^2$

That is $x^2+y^2 = r^2$

This is the equation of the circle with centre $(0,0)$ and radius r.

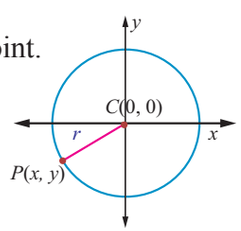


Fig.5.7

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